

M05 Protected Review + Targeted Repair

STOP: open a lesson section only after a genuine saved attempt for that lesson/practical competency.

RULE

Review explains the control and repair direction; it must not erase or replace your first attempt. After repair, close this PDF and complete the exact changed retest task assigned for the weak competency.

DE05-01 Review

What correct work must prove

Row count and field list are explicit; raw copy hash equals source hash; no transformation overwrites raw.

Likely diagnosis

Opening CSV in Excel and saving it back before preserving the original can silently change formatting/types.

Targeted repair

Rebuild only the smallest failing contract for csv properly: parse, inspect, preserve raw. Add one explicit assertion/control that would have caught the failure before publish. Do not copy a finished pipeline.

After repair

Close Review and do changed retest: Receive data/retest/shipments_retest.csv. Save its SHA-256 and row/field controls before doing any cleanup. Create your own byte-identical receipt copy under data/work/retest_received/.

DE05-02 Review

What correct work must prove

Every JSONL line parses; IDs/count reconcile; nested data meaning is not flattened accidentally.

Likely diagnosis

Treating a JSON array file as if each physical line were a full JSON record.

Targeted repair

Rebuild only the smallest failing contract for json and jsonl from zero. Add one explicit assertion/control that would have caught the failure before publish. Do not copy a finished pipeline.

After repair

Close Review and do changed retest: Convert data/retest/shipments_sample.json into JSONL one object per line, then reconstruct the list. Prove shipment_id count/set is identical.

DE05-03 Review

What correct work must prove

Same logical run keys produce same intended paths; no personal absolute path is required.

Likely diagnosis

Hard-coding C:/Users/.../Downloads makes the pipeline non-portable.

Targeted repair

Rebuild only the smallest failing contract for paths, folders and deterministic batch files. Add one explicit assertion/control that would have caught the failure before publish. Do not copy a finished pipeline.

After repair

Close Review and do changed retest: For source=shipments and run_date=2026-08-15, derive landing/accepted/rejected paths twice from project root/config. Prove identical inputs produce identical paths and no C:/Users/... dependency appears.

DE05-04 Review

What correct work must prove

Profile is based on observed source; row/key/null/type controls are saved separately from cleaning.

Likely diagnosis

Cleaning while profiling destroys evidence of what actually arrived.

Targeted repair

Rebuild only the smallest failing contract for pandas profiling before transformation. Add one explicit assertion/control that would have caught the failure before publish. Do not copy a finished pipeline.

After repair

Close Review and do changed retest: Profile shipments_retest.csv before cleaning. Save rows, columns, dtypes, missing counts, duplicate shipment_id count and suspicious date/value/quantity observations.

DE05-05 Review

What correct work must prove

All source rows accounted for; rejects retain source fields + reason; no bad row silently disappears.

Likely diagnosis

dropna() or errors="coerce" followed by filtering can hide defects if rejects are not preserved.

Targeted repair

Rebuild only the smallest failing contract for cleaning with pandas and rejected rows. Add one explicit assertion/control that would have caught the failure before publish. Do not copy a finished pipeline.

After repair

Close Review and do changed retest: Split shipments_retest.csv into accepted and rejected rows with explicit reason codes. Prove $\text{input_rows} = \text{accepted_rows} + \text{rejected_rows}$ and retain original fields.

DE05-06 Review

What correct work must prove

Pre/post row counts preserved when expected; duplicate one-side key blocks the merge; unmatched count is explicit; grouped total reconciles.

Likely diagnosis

A merge can run and still multiply money if the right-side key is not unique.

Targeted repair

Rebuild only the smallest failing contract for pandas merge, cardinality and aggregation. Add one explicit assertion/control that would have caught the failure before publish. Do not copy a finished pipeline.

After repair

Close Review and do changed retest: Attempt many-to-one merge of shipment rows to carriers_retest.csv. Detect the deliberate duplicate carrier key before trusting totals, then separately report unmatched carrier IDs. Do not hide either issue with drop_duplicates().

DE05-07 Review

What correct work must prove

Good data passes; each controlled defect is detected; validation report identifies row/key/rule; publish is blocked on critical failure.

Likely diagnosis

Using type hints alone and assuming runtime values are validated automatically.

Targeted repair

Rebuild only the smallest failing contract for data validation: record, batch and relationship rules. Add one explicit assertion/control that would have caught the failure before publish. Do not copy a finished pipeline.

After repair

Close Review and do changed retest: Create ShipmentRecord validation plus batch uniqueness and carrier relationship checks. Prove the packaged invalid date, zero units, negative unit value, missing carrier, duplicate shipment ID and invalid status are surfaced by named rules.

DE05-08 Review

What correct work must prove

200 payload shape is understood; 404 is not parsed as valid orders; 500 is classified as server failure.

Likely diagnosis

Calling response.json() and using the result before checking status/contract.

Targeted repair

Rebuild only the smallest failing contract for http and rest apis from zero. Add one explicit assertion/control that would have caught the failure before publish. Do not copy a finished pipeline.

After repair

Close Review and do changed retest: Start the bundled local API. Call /health, /shipments?page=1&per_page=3, /missing and /shipments?fail=1. Record method/path/query/status and classify continue/stop/retry behavior.

DE05-09 Review

What correct work must prove

Timeout configured; non-2xx does not enter normal processing; payload contract failures are distinct from HTTP failures.

Likely diagnosis

except Exception: return [] turns outages into false "zero records" success.

Targeted repair

Rebuild only the smallest failing contract for calling apis safely with requests. Add one explicit assertion/control that would have caught the failure before publish. Do not copy a finished pipeline.

After repair

Close Review and do changed retest: Implement a safe page fetch for /shipments with timeout, status check and required payload keys. Then call /shipments?bad_payload=1 and prove it fails as a payload-contract error rather than returning zero rows.

DE05-10 Review

What correct work must prove

All expected IDs fetched once; token absent from source/log; auth failure stops clearly; pagination termination is deterministic.

Likely diagnosis

Hard-coding bearer token into Python or logging full Authorization headers.

Targeted repair

Rebuild only the smallest failing contract for pagination and authentication without leaking secrets. Add one explicit assertion/control that would have caught the failure before publish. Do not copy a finished pipeline.

After repair

Close Review and do changed retest: Fetch all /shipments pages using API_TOKEN from the environment. Repeat with per_page=2 and per_page=4; the ordered unique shipment ID set must match. Prove the token is absent from code/log evidence.

DE05-11 Review

What correct work must prove

Retries bounded; 429/5xx policy explicit; partial completion state distinguishable from full success; no infinite loop.

Likely diagnosis

Blind retry around all exceptions can hammer a service and conceal permanent bad data.

Targeted repair

Rebuild only the smallest failing contract for retries, rate limits and partial failure. Add one explicit assertion/control that would have caught the failure before publish. Do not copy a finished pipeline.

After repair

Close Review and do changed retest: Use `/shipments?rate_limit_once=1` to prove one bounded recovery, then `/shipments?fail=1` to prove a permanent failure is not mislabeled full success. Save attempt count and last-safe state.

DE05-12 Review

What correct work must prove

No credential in code; connection works; row/key/totals reconcile; failure rolls back; rerun policy is explicit.

Likely diagnosis

Embedding postgres password in source or assuming commit happened because `execute()` returned.

Targeted repair

Rebuild only the smallest failing contract for python + postgresql with sqlalchemy. Add one explicit assertion/control that would have caught the failure before publish. Do not copy a finished pipeline.

After repair

Close Review and do changed retest: Load the accepted retest shipment batch to a separate `m05_retest` table/run_id using SQLAlchemy and the existing M03 PostgreSQL install. Inject one transaction failure and prove rollback leaves no half-published batch.

DE05-13 Review

What correct work must prove

Same code runs with two configs; missing required setting fails clearly; secrets are excluded from Git/log evidence.

Likely diagnosis

A `.env` file is convenient locally but is not safe to commit just because it is hidden in Explorer.

Targeted repair

Rebuild only the smallest failing contract for configuration, environment variables and secret hygiene. Add one explicit assertion/control that would have caught the failure before publish. Do not copy a finished pipeline.

After repair

Close Review and do changed retest: Create a second non-secret config profile for shipments with a different output folder/batch size. Override the fake training token through the environment without editing pipeline logic; scan evidence for secret leakage.

DE05-14 Review

What correct work must prove

Tests fail before repair and pass after; logs identify run and counts; functions have clear responsibilities; failures propagate.

Likely diagnosis

A green test suite that never checks bad input/rerun behavior gives false confidence.

Targeted repair

Rebuild only the smallest failing contract for logging, testing and reusable pipeline code. Add one explicit assertion/control that would have caught the failure before publish. Do not copy a finished pipeline.

After repair

Close Review and do changed retest: Write a regression test for one shipment validation/logging bug, capture the failing test first, repair code, then capture the passing test. Log `run_id`, source, accepted/rejected counts and final status.

DE05-15 Review

What correct work must prove

PASS only when all integrated pipeline contracts are evidenced: raw/source receipt preserved; accepted + rejected reconcile to input; no silent bad-data publish; rerun creates no duplicate publish; transient failure follows bounded recovery; permanent/partial failure is not labeled full success; no secret appears in code/log/evidence; output or PostgreSQL state reconciles; tests/logs support the run; and another learner could reproduce the run from README/config. Any unresolved critical contract => TARGETED REPAIR.

Likely diagnosis

Treating "script exited 0" as sufficient without proving data correctness, state and recovery.

Targeted repair

Rebuild only the smallest failing contract for integrated python data-engineering pipeline. Add one explicit assertion/control that would have caught the failure before publish. Do not copy a finished pipeline.

After repair

Close Review and do changed retest: Run the complete changed shipments pipeline from raw receipt + local API through validation/quarantine, transformation, bounded retry state, optional PostgreSQL load, logs/tests and reconciliation. Preserve the original practical attempt; this fresh run is separate evidence.